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0.1 Cantor Set

Define operations on $\mathcal{P}(\mathbb{R})$ as: for $A, B \subset \mathbb{R}$ and $c \in \mathbb{R}$,

$$A + B \stackrel{\text{def}}{=} \{a + b \mid a \in A, b \in B\}$$

$$cA \stackrel{\text{def}}{=} \{ca \mid a \in A\}$$

If $A = \{\alpha\}$ is a singleton, denote $A + B$ as $\alpha + B$.

Definition 0.1.1. Let $C_0 \stackrel{\text{def}}{=} [0, 1]$ and $C_n \stackrel{\text{def}}{=} \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$, for $n \geq 1$.

An intersection $C = \bigcap_{n=1}^{\infty} C_n$ is called a *Cantor Set*.

Cantor set is a remarkable object in the Real numbers set.

In particular, Cantor Set is *Uncountable, Compact, Perfect, Nowhere Dense*, and has *Measure zero*.

Moreover, sum of two Cantor sets is $[0, 2]$.

And, total length (or measure) of C is zero, because:

$$\#C_0 = 1,$$

$$\#C_1 = \frac{2}{3} = 1 - \frac{1}{3},$$

$$\#C_2 = 1 - \frac{1}{3} - \frac{1}{3^2} \cdot 2,$$

⋮

$$\#C_n = 1 - \sum_{i=0}^n \frac{2^i}{3^{i+1}} = 1 - \frac{1}{3} \cdot \frac{1 - \left(\frac{2}{3}\right)^{n+1}}{1 - \frac{2}{3}} = 1 - \left(1 - \left(\frac{2}{3}\right)^{n+1}\right) = \left(\frac{2}{3}\right)^{n+1}.$$

0.2 Ternary Expansion

Definition 0.2.1. Given $x \in (0,1)$, a sequence $\{n_i\}_{i=1}^{\infty}$ in $\{0,1,2\}$ is called a *ternary expansion* of x if:

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

Theorem 0.2.2. Given $x \in (0,1)$, a ternary expansion of x exists.

Proof. Let $x \in (0,1)$ be given. We construct the sequence $\{n_i\}$ inductively as follows:
First, choose $n_1 \in \{0,1,2\}$ to be the largest integer such that

$$\frac{n_1}{3} < x$$

Next, for chosen $n_1, n_2, \dots, n_{k-1}$, we choose the largest digit $n_k \in \{0,1,2\}$ such that

$$\frac{n_1}{3} + \frac{n_2}{3^2} + \dots + \frac{n_{k-1}}{3^{k-1}} + \frac{n_k}{3^k} < x$$

From this construction of $\{n_i\}$, it follows that for any $k \in \mathbb{N}$,

$$\sum_{i=1}^k \frac{n_i}{3^i} < x \leq \sum_{i=1}^k \frac{n_i}{3^i} + \frac{1}{3^k}$$

Note that the partial sum $\sum_{i=1}^k \frac{n_i}{3^i}$ increases monotonically and is bounded by 1, thus it converges.
Taking the limit as $k \rightarrow \infty$ in the above inequality, we obtain:

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} \leq x \leq \sum_{i=1}^{\infty} \frac{n_i}{3^i} + \lim_{k \rightarrow \infty} \frac{1}{3^k}$$

Since $\lim_{k \rightarrow \infty} \frac{1}{3^k} = 0$, we have

$$x = \sum_{i=1}^{\infty} \frac{n_i}{3^i}$$

which gives the desired result. □

Note: the ternary expansion described in the existence theorem is called the *First-form* ternary expansion.

Theorem 0.2.3. Given $x \in (0, 1)$, the ternary expansion of x is unique except: for some $m \in \mathbb{N}$ and $0 < a < 3^m$,

$$x = \frac{a}{3^m}, \text{ irreducible form} \quad (*)$$

Particularly, if x can be represented as $(*)$, it has only two different ternary expansion.

Proof. First, we show that if x is of the form $(*)$, it has only two expansions. If x can be represented as $(*)$, then

$$x = \frac{a}{3^m} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m}$$

where $a_i \in \{0, 1, 2\}$ with $a_m \neq 0$. It is obtained by the Euclidean Algorithm:

$$a = 3^{m-1}a_1 + 3^{m-2}a_2 + \cdots + 3^2a_{m-2} + 3a_{m-1} + a_m$$

(If $a_m = 0$, then x can be divided by 3, it is contradiction to irreducibility.) Meanwhile, note that

$$\sum_{k=m+1}^{\infty} \frac{2}{3^k} = \frac{2}{3^{m+1}} \cdot \frac{1}{1 - 1/3} = \frac{2}{3^{m+1}} \cdot \frac{3}{2} = \frac{1}{3^m}.$$

Therefore, we obtain two ternary expansion of x as:

$$x = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m}{3^m} + \sum_{k=m+1}^{\infty} \frac{0}{3^k} = \frac{a_1}{3} + \frac{a_2}{3^2} + \cdots + \frac{a_m - 1}{3^m} + \sum_{k=m+1}^{\infty} \frac{2}{3^k} \quad (**)$$

This proves that such x has two distinct expansions. Meanwhile, suppose that x has two distinct expansions $\{n_i\}, \{r_i\}$. Let $j \in \mathbb{N}$ be the smallest index such that $n_j \neq r_j$. Since $n_i = r_i$ for all $i < j$, we have:

$$0 = \sum_{i=1}^{\infty} \frac{n_i - r_i}{3^i} = \frac{n_j - r_j}{3^j} + \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \quad (***)$$

Since $n_j \neq r_j$ and

$$\left| \sum_{i=j+1}^{\infty} \frac{n_i - r_i}{3^i} \right| \leq \sum_{i=j+1}^{\infty} \frac{2}{3^i} = \frac{1}{3^j}$$

the $(***)$ holds if and only if

$$n_j - r_j = 1, n_i - r_i = -2 \text{ for all } i \geq j+1 \text{ or } n_j - r_j = -1, n_i - r_i = 2 \text{ for all } i \geq j+1$$

But this equivalent to x can be represented in the form of $(*)$, therefore we proved the Uniqueness. $\square$

Note: the ternary expansion is of the form $(*)$ has two expansions. Clearly, the left side in $(**)$ is a first form. The right side form is called a *Second form*.

0.3 Ternary representation of Cantor Set

Theorem 0.3.1. $C = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 2\} \right\}$ where C is the Cantor set.

Proof. Since the existence of ternary expansion,

$$C_0 = [0, 1] = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 1, 2\} \right\}$$

Observe that

$$\begin{aligned} C_1 &= \frac{C_0}{3} \cup \left(\frac{2}{3} + \frac{C_0}{3} \right) = \left\{ \frac{0}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0, 1, 2\} \right\} \cup \left\{ \frac{2}{3^1} + \sum_{i=2}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=2}^{\infty} \subseteq \{0, 1, 2\} \right\} \\ &\vdots \\ C_k &= \bigcup_{n_1, \dots, n_k \in \{0, 2\}} \left\{ \frac{n_1}{3^1} + \frac{n_2}{3^2} + \dots + \frac{n_k}{3^k} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=k+1}^{\infty} \subseteq \{0, 1, 2\} \right\} \\ &= \left\{ \sum_{i=1}^k \frac{n_i}{3^i} + \sum_{i=k+1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\}_{i=1}^k \subseteq \{0, 2\} \text{ and } \{n_i\}_{i=k+1}^{\infty} \subseteq \{0, 1, 2\} \right\} \end{aligned}$$

Finally,

$$C \stackrel{\text{def}}{=} \bigcap_{k \in \mathbb{N}} C_k = \left\{ \sum_{i=1}^{\infty} \frac{n_i}{3^i} \mid \{n_i\} \subseteq \{0, 2\} \right\}$$

□

Theorem 0.3.2. $C + C = [0, 2]$ where C is the Cantor set.

Proof. Since $C \subseteq [0, 1]$, $C + C \subseteq [0, 2]$ is clear. Let $x \in [0, 2]$ be given.

Trivially $\frac{x}{2} \in [0, 1]$. Therefore, there exists a ternary expansion $\{n_i\} \subseteq \{0, 1, 2\}$ s.t.

$$\sum_{i=1}^{\infty} \frac{n_i}{3^i} = \frac{x}{2}$$

Use this fact, we can write that

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i}$$

Define

$$a_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 0 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases} \quad \text{and} \quad b_i = \begin{cases} 0 & \text{if } n_i = 0 \\ 2 & \text{if } n_i = 1 \\ 2 & \text{if } n_i = 2 \end{cases}$$

Then, $2n_i = a_i + b_i$ for all $i \geq 1$. And,

$$x = \sum_{i=1}^{\infty} \frac{2n_i}{3^i} = \sum_{i=1}^{\infty} \frac{a_i}{3^i} + \sum_{i=1}^{\infty} \frac{b_i}{3^i} \in C + C$$

□

A

Prop. Let C be a Cantor set, and χ_C be a characteristic function defined on $[0,1]$

- 1). The map χ_C is continuous on $[0,1] \setminus C$.
- 2). The map χ_C is discontinuous at every point on C .
- 3). The map χ_C is differentiable on $[0,1] \setminus C$.
- 4). The map χ_C is Riemann-Integrable on $[0,1]$.

Proof. All C_n ($n \in \mathbb{N}$) are closed, so $C = \bigcap_{n \in \mathbb{N}} C_n$ is also closed set.

Therefore, given $x \in [0,1] \setminus C$, there exists $r > 0$ such that $(x-r, x+r) \subseteq [0,1] \setminus C$.

Now, for all $t \in (x-r, x+r)$, $|\chi_C(x) - \chi_C(t)| = |0 - 0| = 0$, so χ_C is continuous.

Similarly, 3) can be proved. To show the second assertion, let $x \in C$ be given

Fix $\varepsilon = \frac{1}{2}$. Given $\delta > 0$, choose $n \in \mathbb{N}$ such that $1/3^n < \delta$. Denote $B = (x-\delta, x+\delta) \cap [0,1]$.

By definition, $x \in C_n$ and there exists $t \in B \setminus C_n$, because:

If no such t exists, that is, $B \setminus C_n = \emptyset$, it equals to $B \subseteq C_n$.

But, the C_n is union of disjoint closed intervals of length $1/3^n$, it contradicts $1/3^n < \delta$.

Thus, $C \subseteq C_n$ implies $t \in B \setminus C_n \subseteq B \setminus C$, it implies

$$|f(x) - f(t)| = |1 - 0| = 1 > \frac{1}{2} = \varepsilon.$$

Consequently, χ_C is not continuous at $x \in C$.

To show the last assertion, observe that

$$0 \leq \int_0^1 \chi_C(x) dx \leq \int_0^1 \chi_{C_n}(x) dx \leq \left(\frac{2}{3}\right)^n \cdot 3, \text{ because:}$$

Let $P_n := \{a/3^n \mid a \in \mathbb{Z}, 0 \leq a \leq 3^n\}$ be a partition on $[0,1]$, then the upper sum

$$\begin{aligned} U(f, P_n) &= \sum_{i=1}^{3^n} M_i \cdot \frac{1}{3^n} \quad \text{where } M_i = \sup \{ \chi_C(x) \mid \frac{i-1}{3^n} \leq x \leq \frac{i}{3^n} \} \\ &\leq \frac{2^n \cdot 3}{3^n} \end{aligned}$$

so, $\int_0^1 \chi_C \leq U(f, P_n)$ by the definition, we obtain the inequality.

Consequently, letting $n \rightarrow \infty$, we obtain the result. $\square$

Prop. Let C be the Cantor set, and let f be a continuous at except C .
Then, f is Riemann-integrable.